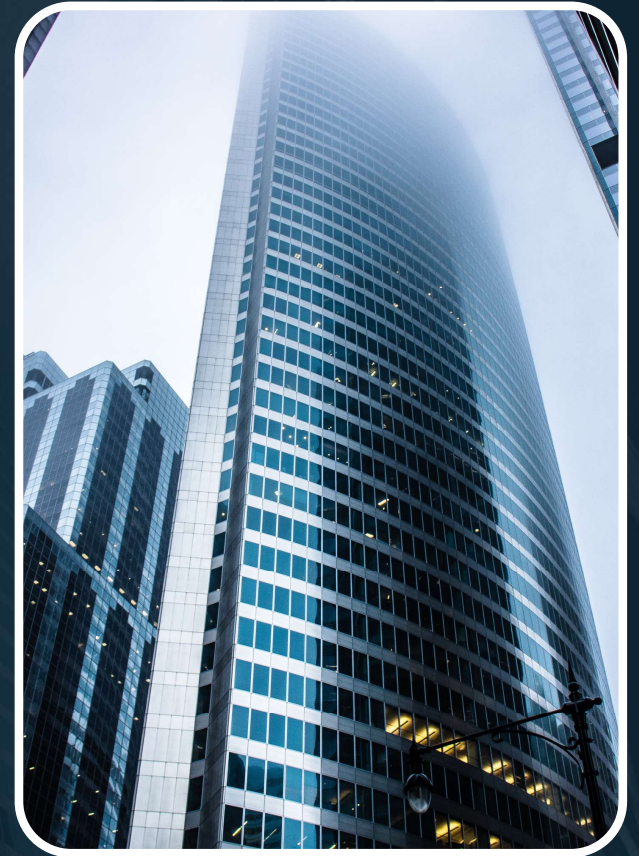

Matsuo Lab,
University of Tokyo, Japan

BUSINESS PROPOSAL

“AI-Driven Customer
Churn
Reduction Strategy for
Telecom Operator”



Presented by : Elewi Ebenezer Mayowa

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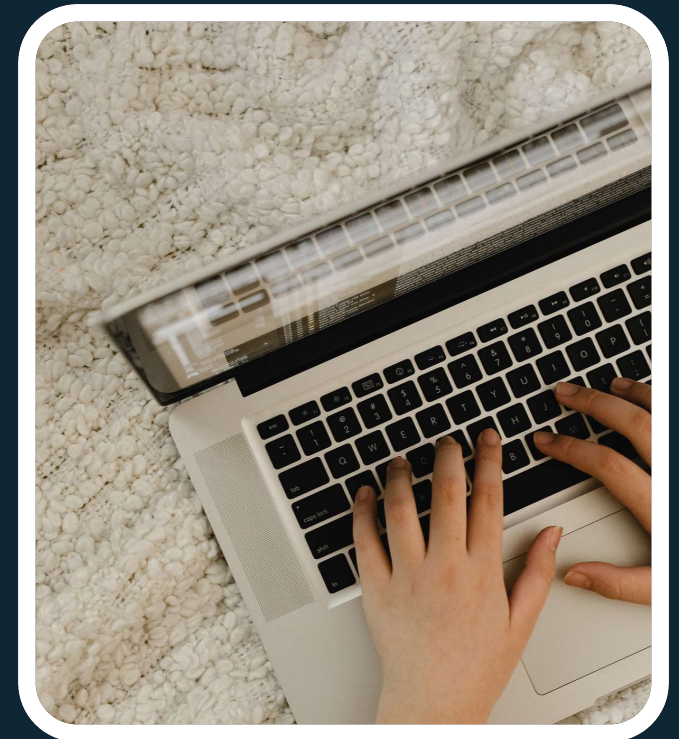


PROBLEM STATEMENT

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The telecom company is losing approximately half of its customer base (49.6% Churn rate), which is extraordinarily high for a telecom company. The implication is that, out of every 100 customers acquired, nearly 50 leave.

Additionally, there's a risk of revenue leakage and increased acquisition costs.



MARKET ANALYSIS

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According to research from Harvard Business Review, acquiring a new customer costs 5 times more than retaining an existing one who has already chosen you, already integrated your product into their workflow, and already proved they see value in what you're selling (Baird Hall, Feb 20, 2026)

Retaining customers is up to five times cheaper than acquiring new ones, making retention strategies essential for sustainable profitability. (Mark Camp, July 4, 2025)

A 5% Increase in customer retention can deliver a 25% increase in profits.

DATASET OVERVIEW

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100,000 customers

100 variables: Behavioral, demographic,
device, and revenue attributes

Mean revenue = \$1031

Median revenue = \$804

Maximum revenue = \$27,321

The large gap between mean and
median indicates a right-skewed
distribution.

Interpretation:

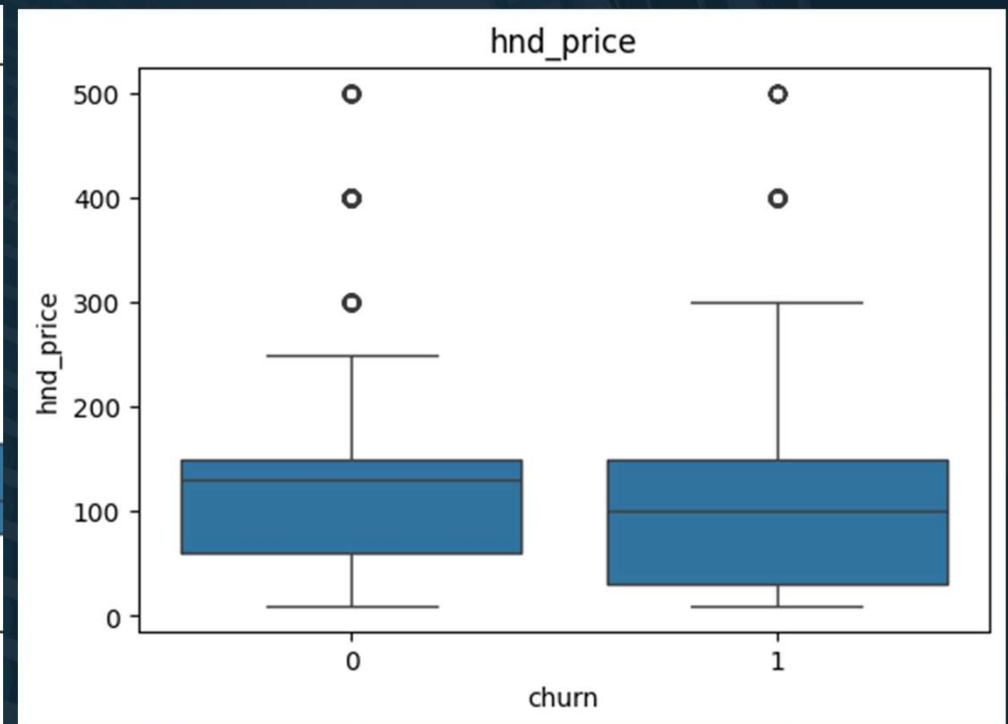
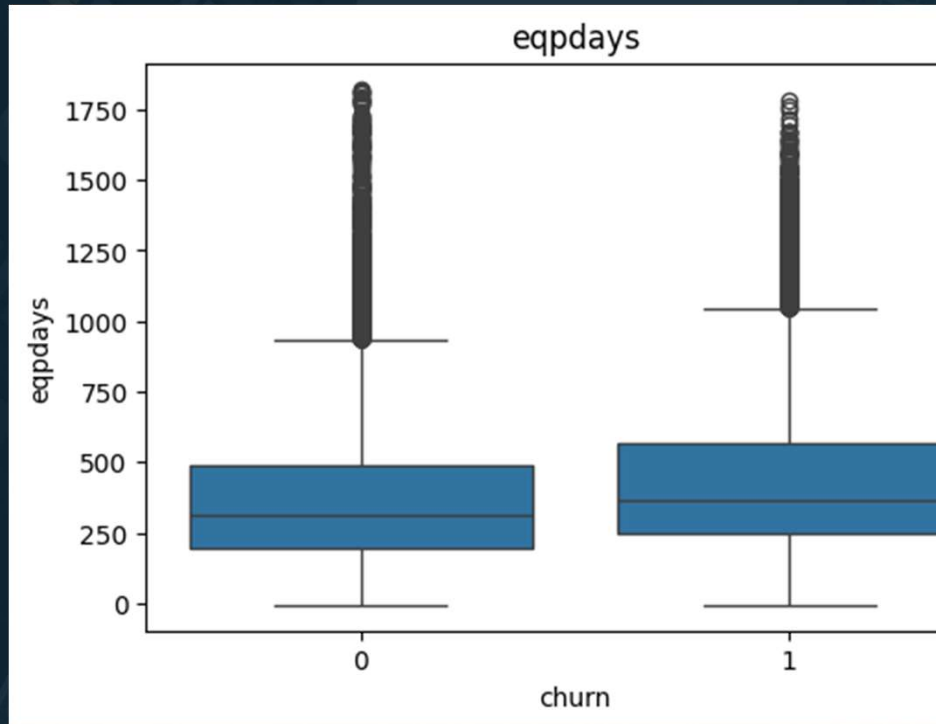
Most customers generate moderate revenue.

A small group generates extremely high revenue.



CHURN OVERVIEW

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Insight #1: Device Age is the strongest churn signal:

eqpdays: +0.113 correlation with churn

hnd_price: -0.103 correlation

Insight #2: Service Quality is probably not the main churn driver.

Churners actually report fewer service issues and customer-care interactions.

Insight #3: High Usage Customers Tend to Stay. Interpretation:

The more customers use the service, the less likely they are to leave.

This is extremely common in telecoms.

High engagement = lower churn.

FEATURE ENGINEERING

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- Missing value handling
- One-hot encoding
- Column removal
- Preprocessing Pipeline

Three Models were trained and tested:

1. Logistic Regression
2. Random Forest
3. Gradient Boosting

Logistic Regression was trained as the baseline model and achieved an ROC-AUC of 0.605. However, two limitations necessitated training other advanced models. The limitations are:

- Logistic Regression assumes mostly linear relationships
- It is unable to fully capture the complexity of customer behavior

MODEL COMPARISON

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Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.578	0.577	0.557	0.567	0.605
Random Forest	0.611	0.599	0.65	0.624	0.664
Gradient Boosting	0.625	0.614	0.654	0.633	0.677

Best Model:

Gradient Boosting

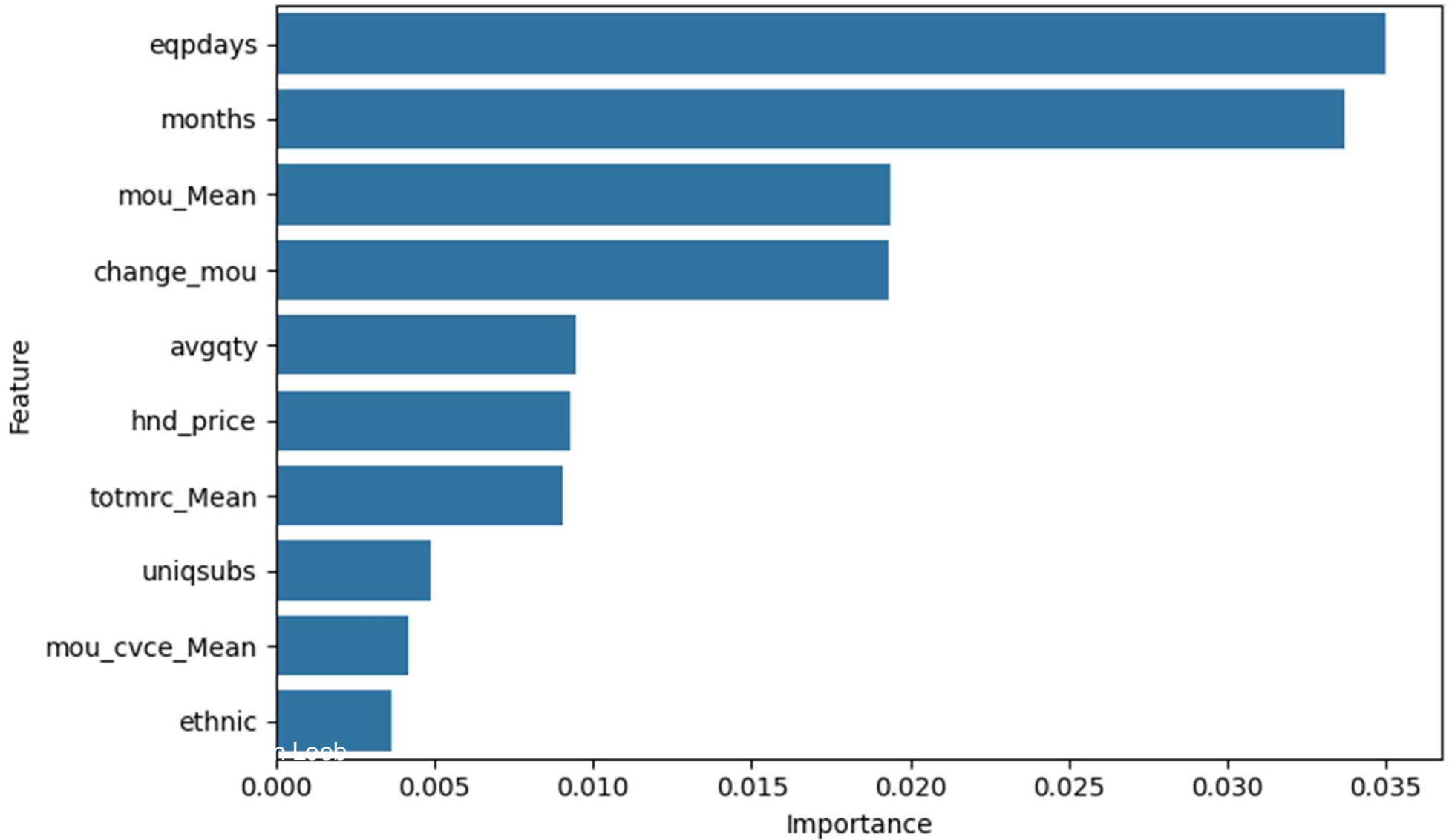
Accuracy = 62.5%

Recall = 65.4%

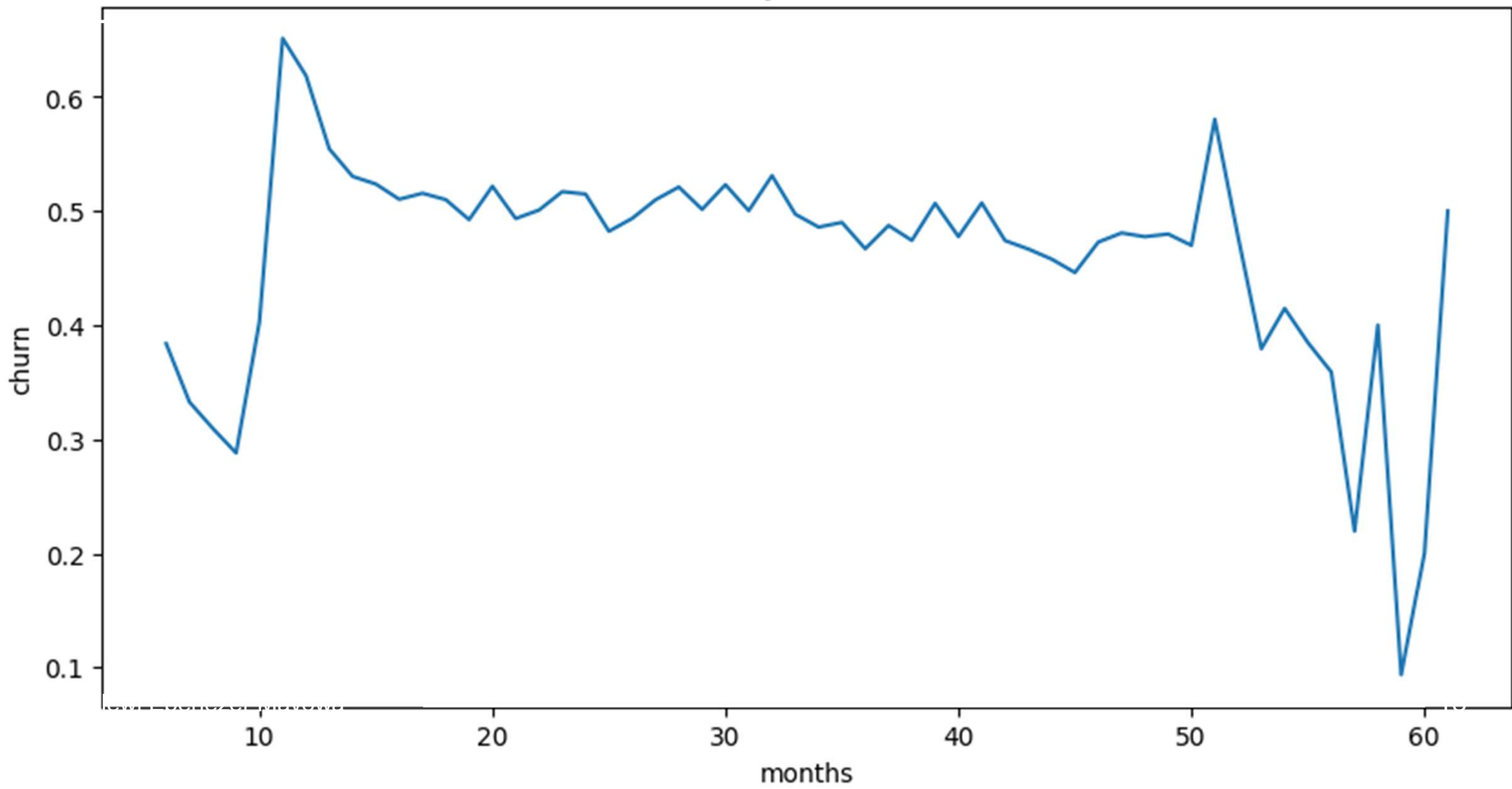
ROC-AUC = 67.7%

FEATURE IMPORTANCE

Top Drivers of Customer Churn



Churn Rate by Customer Tenure



FEATURE IMPORTANCE

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Major Business Insights:

#1 - Device Lifecycle is the strongest churn indicator, i.e., customers become increasingly vulnerable to churn as their devices age.

#2 - Customer Tenure matters more than revenue. Possibly, tenure windows may be high-risk churn periods.

#3 - The model heavily uses: mou_Mean, change_mou, and avgqty. This showcases that declining usage is an early warning sign of churn.

#4 -As indicated by the Feature Importance analysis, the root causes identified as the predictors of churn are:

- 1.Aging devices
- 2.Customer tenure stage
- 3.Declining service usage
- 4.Lower handset value

ESTIMATED BUSINESS IMPACT

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Revenue Impact Calculation

From the dataset:

- Customers = 100,000
- Churn Rate = 49.56%
- Churned Customers $\approx 49,562$
- Average Revenue per Customer $\approx \$1,031.92$

If the Program Reduces Churn by 5%:

Customers retained: $49562 * 0.05$

$\sim 2,478$

Revenue preserved: $2478 * 1031.92$

$\sim \$2.56 \text{ Mn}$

If the Program Reduces Churn by 10%:

Customers retained: $49562 * 0.10$

$\sim 4,956$

Revenue preserved: $4956 * 1031.92$

$\sim \$5.11 \text{ Mn}$

If the Program Reduces Churn by 15%:

Customers retained: $49562 * 0.15$

$\sim 7,434$

Revenue preserved: $7434 * 1031.92$

$\sim \$7.67 \text{ Mn}$

BUSINESS PROPOSAL

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AI-Powered Churn Prevention Program

Deploy a churn prediction system that scores customers monthly and automatically triggers retention campaigns for high-risk subscribers.

Quantified Impact Estimate:

- customers saved
- retained revenue
- reduced acquisition costs

Monthly:

- Score customer churn risk.
- Identify high-risk subscribers.
- Trigger personalized interventions.
- Prioritize customers with aging devices and declining

THANK YOU

